



Bespoke Bathroom Architecture.

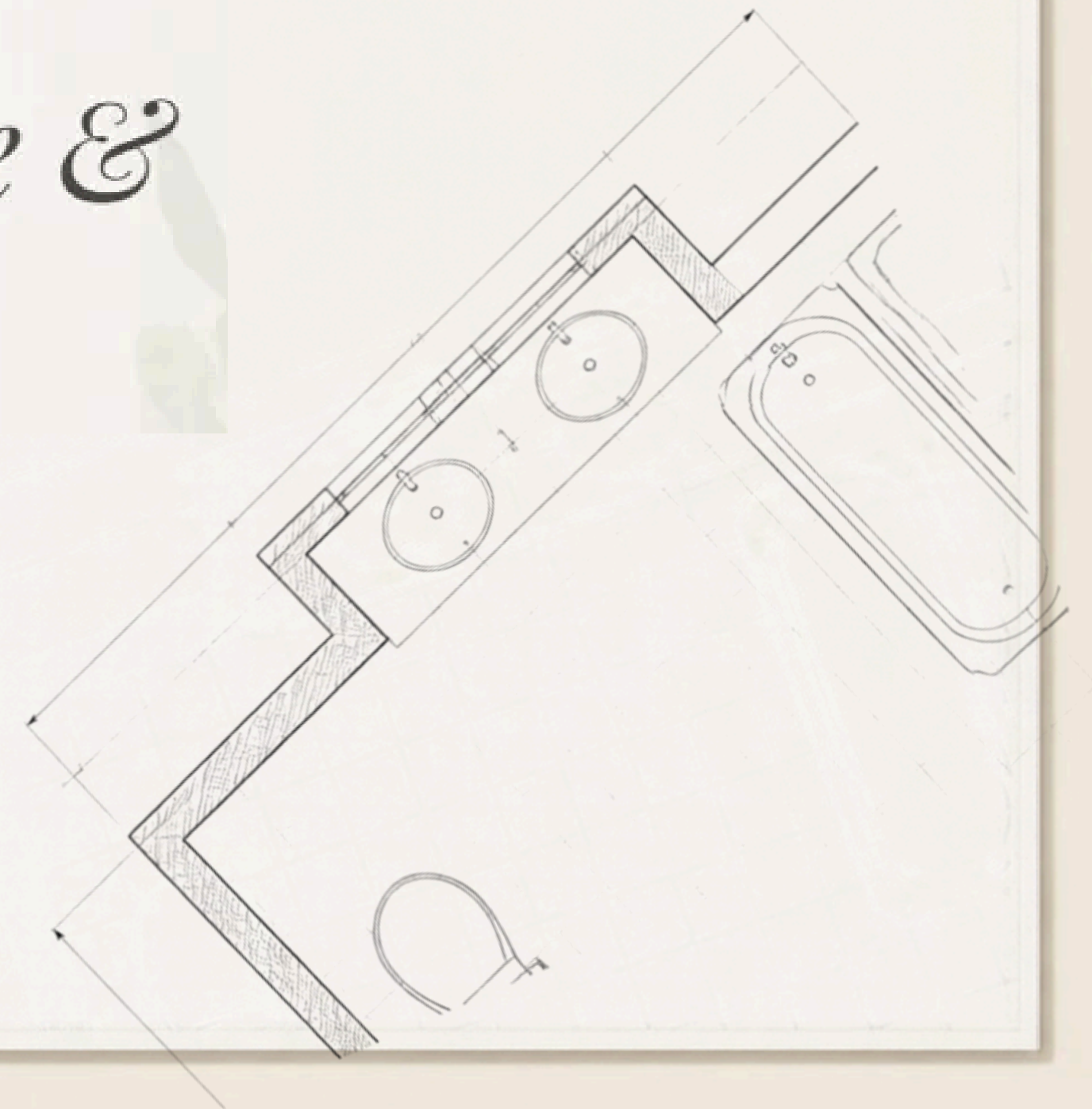
*Engineered for your Space &
Budget.*

TRACK 1: AI Bathroom Designer & Planner

PLUMBLINE

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Solution

Real world problem

Designs fail after purchase or install
Fixtures misfit, door interference, budget overruns

Selection and layout disconnected
No tool links product clearances to room

Generic LLMs hallucinate coordinates and fake savings

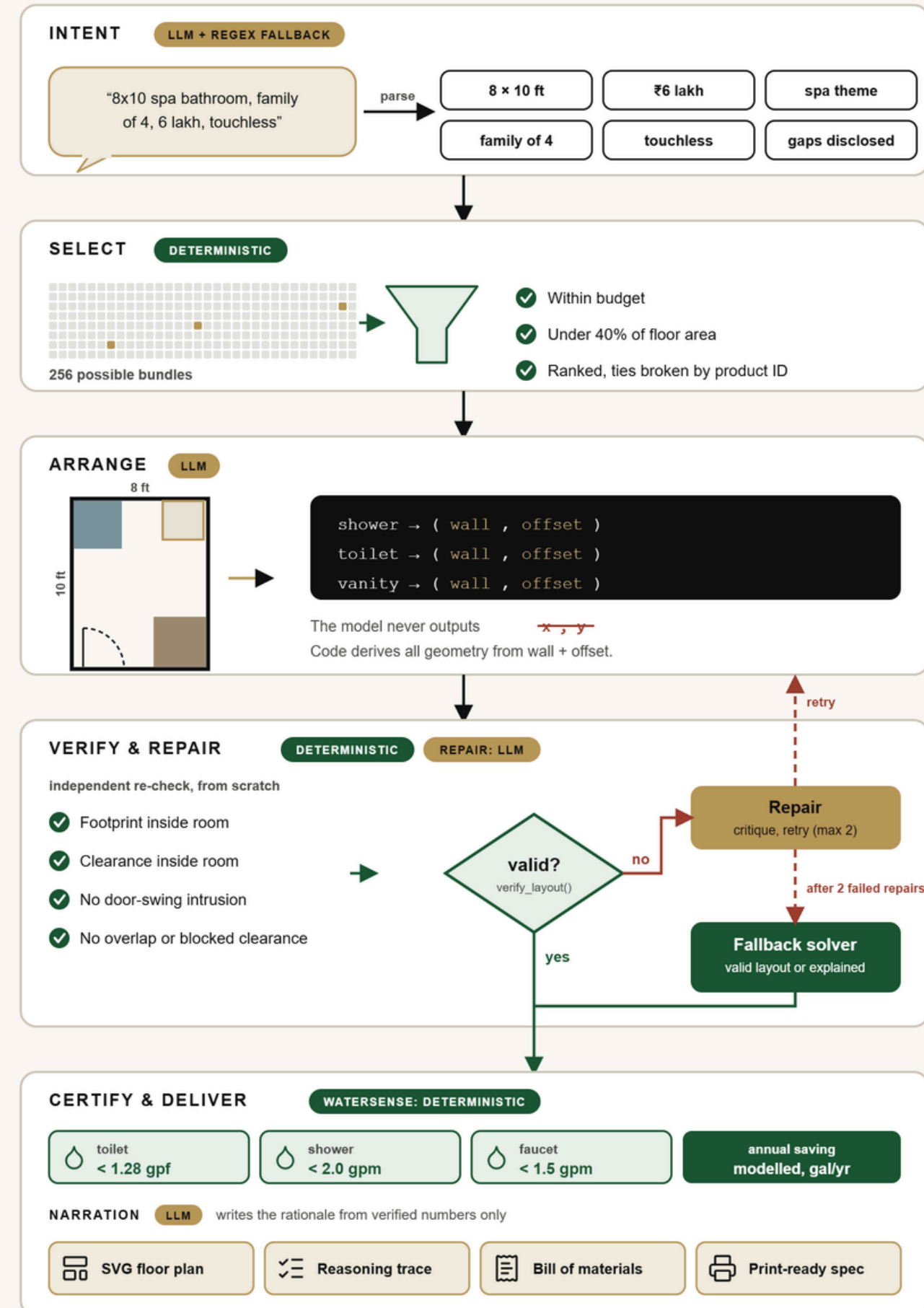
Innovation & uniqueness

Constrained action space: model outputs only (wall, offset), never raw x/y

Verifier-as-critic: violations generate critiques model must resolve

Guaranteed output: fallback solver ensures valid plan, no unverified layouts

Grounded narration: explanations only use verified numeric data



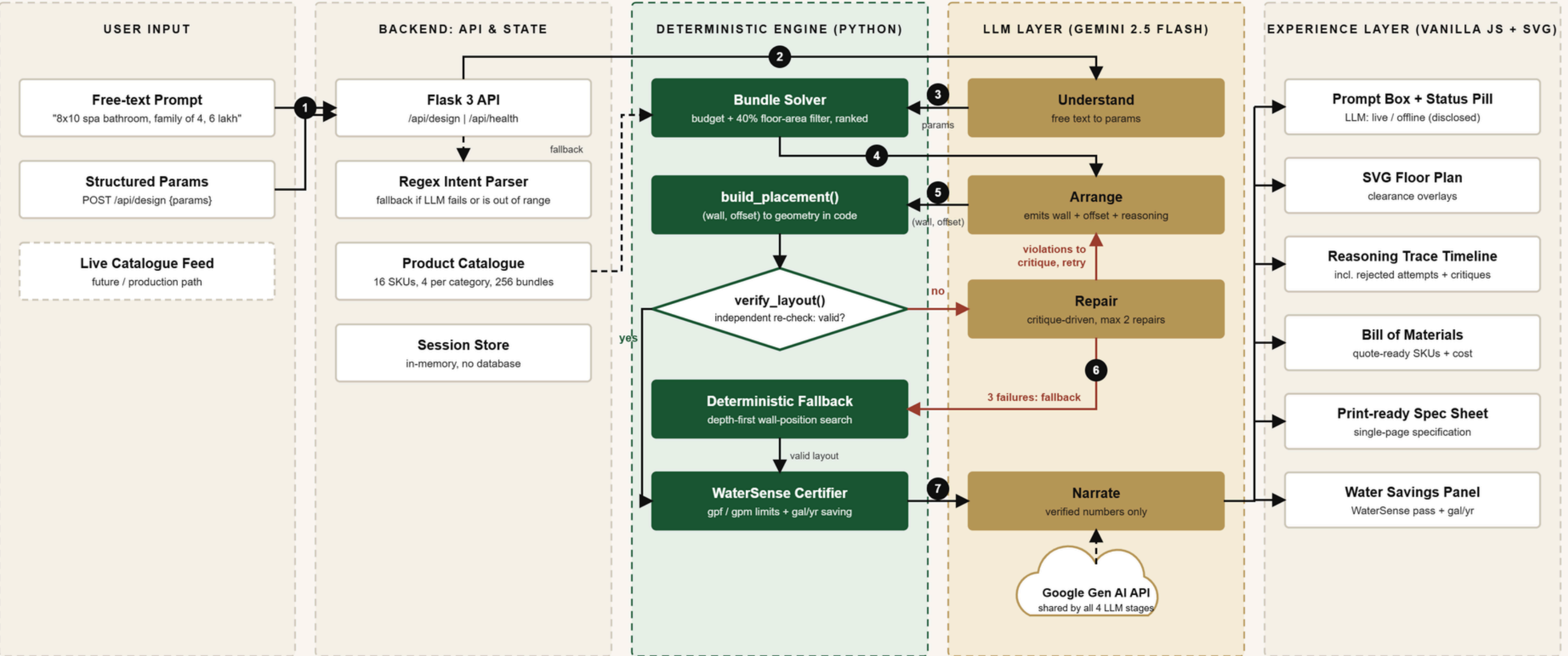
Technical Approach



TECH STACK



Plumblane: System Architecture



The LLM proposes. Geometry disposes.

- Infrastructure / UI
- LLM stages
- Deterministic logic
- Future path

Innovation Pitch



TYPICAL APPROACH		PLUMBLINE
LAYOUT	Model outputs raw coordinates	Model picks (wall, offset); code derives geometry
ROOM FIT	Catalogues without room awareness	Every layout verified for clearance, door swing, overlap
FAILURES	Silent or overlapping	Critique loop + guaranteed fallback
WATER CLAIMS	Invented savings	WaterSense limits as hard constraints
SPEED	CAD needs a professional and hours	One sentence, 1.8 ms p50 offline
TRANSPARENCY	Black box	Full reasoning trace

MEASURED RESULTS

22	passing tests, incl. 4 with stubbed model — no API key needed
1.8 ms	p50 offline response
27,886	gal/yr saved — 8×10 ft, family of 4
45.6%	water reduction vs legacy baseline
3 / 3	WaterSense certified per design

BUILT FOR A MANUFACTURER

- Live catalogue** — `load_catalog()` swaps to PIM API. Solver and verifier don't move.
- Regional codes** — Clearances are data not logic. IS 1172 → IBC → ADA = one config swap.
- Multi-room** — `solve_bathroom_bundle()` is stateless. Parallelise per room trivially.
- Dealer quotes** — `/api/design` returns SKUs + prices + verified plan. Quote-ready now.
- Layout A/B** — Sample at higher temperature. Verifier makes aggressive sampling safe.